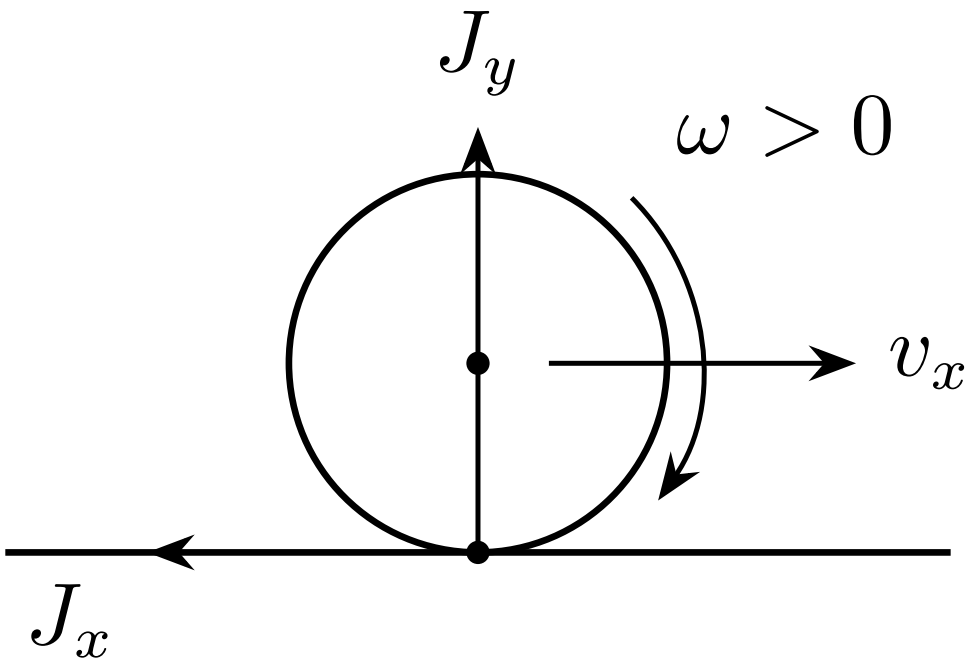

顺时针取正，$u = v_x - R\omega$